State Space Methods for Flight Vehicles

AE 695

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Control Theory: Empirical vs Analytical Methods

Goal of Control Engineering

To design and build real physical systems to perform given tasks.

- **Example**: Design and install a heat exchanger to control the temperature and humidity of a large building.
- **Empirical method**: Determination of required capacity of exchanger from total volume and geographical location of the building.
- Replacement of the exchanger if not powerful enough to control the building's environment.
- Relies heavily on the experience and repeated experimentation.
- Inadequate without the experience or infeasible experimentation because of high cost or risk.
- Examples of infeasible experimentation:
  - Task of sending astronauts to the moon and bringing them back safely.
  - Design of fusion control in nuclear power plants.



Control Theory: Empirical vs Analytical Methods

- **Analytical method**: Systematic procedures for designing systems and reducing the likelihood of designing flawed or disastrous systems.
- Modeling, setting up mathematical equations, analysis, and design.
- Do we need repeated experimentation for analytical methods, similar to empirical ones?
- Yes, but using numerical simulations before the actual application.
- With the adequately chosen model, the resemblance of system performance with that predicted by analytical designs or computer simulations.
- Improvement in performance of physical systems by fine adjustments or tunings (discrepancies due to unavoidable inaccuracy in modeling).
- Lengthy testing before the actual operation or mass production.
- Experimentation needed to set up models and experience needed to improve the performance.



Control Theory: System Definition

System

A collection of components that interact with one another and with their environment.

- A set of interconnected or interdependent components or parts that work together to achieve a common goal or purpose.
- Examples
  - ▷ Human beings, mechanical devices, and an electrical switch.
  - ▷ Plants, animals, the atmosphere, the stock market, and the political system.
  - ▷ Aerospace systems like aircraft, helicopters, missiles, avionics, rocket engines, and so on.
- Assumptions always needed for the design and analysis of systems.
- **Physical Model**: An idealized physical system
- A system may be modeled or viewed differently depending on scenarios.



What is a "Systems" View?

Example

A satellite may be considered a point mass object, a rigid body, or a flexible body, depending on the specific study intended.

- Building a model with reasonable accuracy requires gradual removal of idealized assumptions until a point of diminishing return is reached.
- Gain in accuracy of representations is not commensurate with the increased complexity of the computations needed.
- Physical model $\Longrightarrow$ Mathematical model
- A mathematical representation using appropriate physical laws
- Mathematical models may differ depending on the choice of variables, the coordinate system, etc.
- **Aim to achieve high modeling accuracy**
- A particular mathematical model that gives a greater insight into the governing dynamics and behavior of the system is typically deemed viable for a given application.

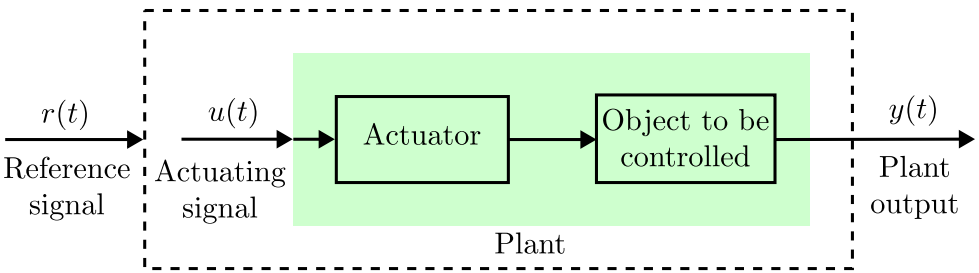


Control System Definition and Performance

Control system

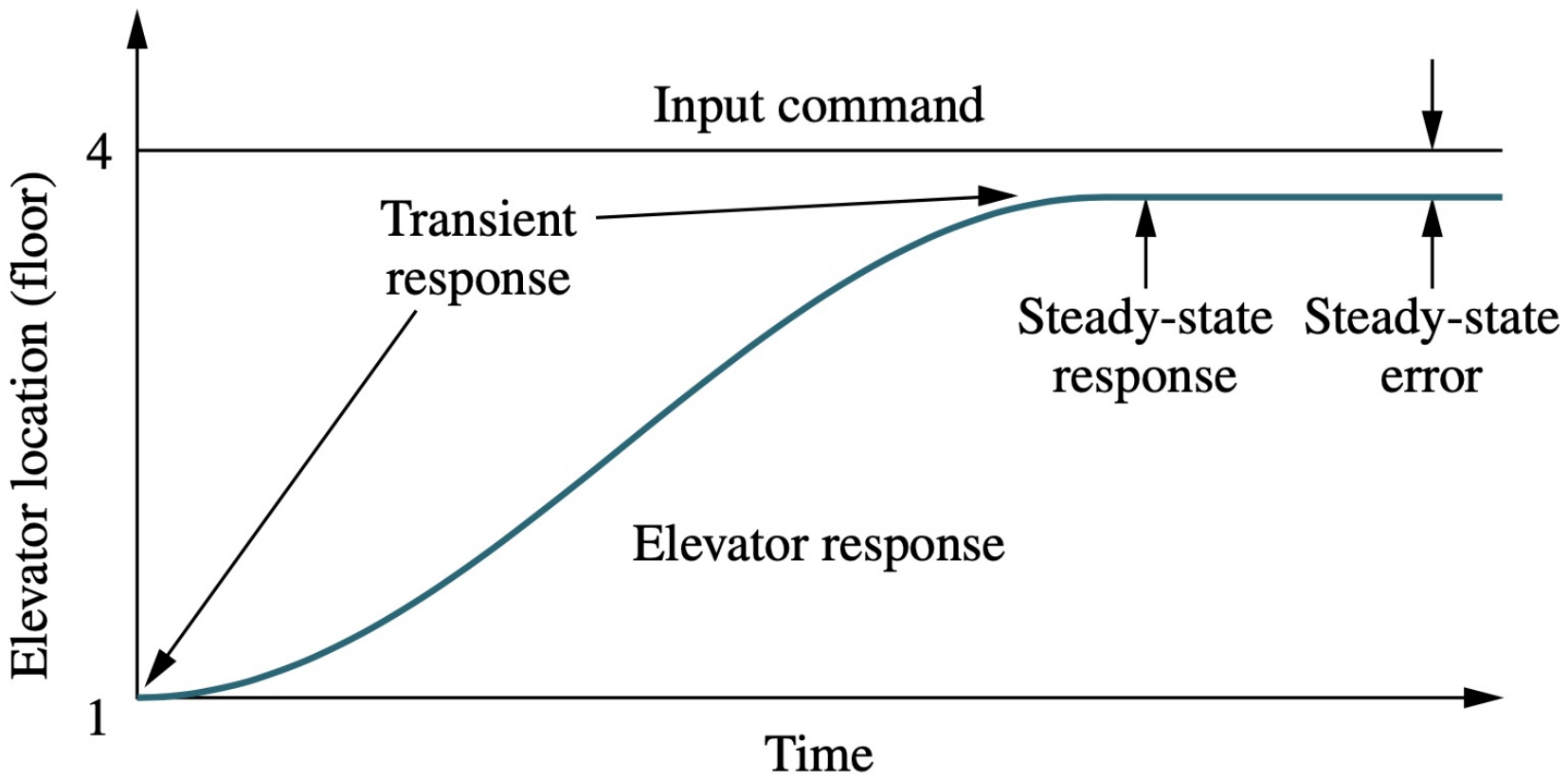
A control system is a collection of components designed to drive a given system plant, with a given input to a desired output.





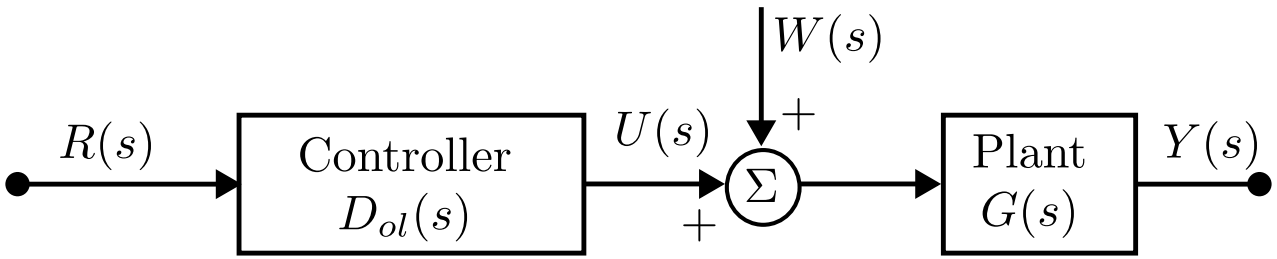
- Primary reasons for control
  - Power amplification
  - Remote control
  - Convenience of input form
  - Compensation for disturbances
- Measures of system's performance
  - Transient response
  - Steady-state error

Control Theory: Performance of Control System



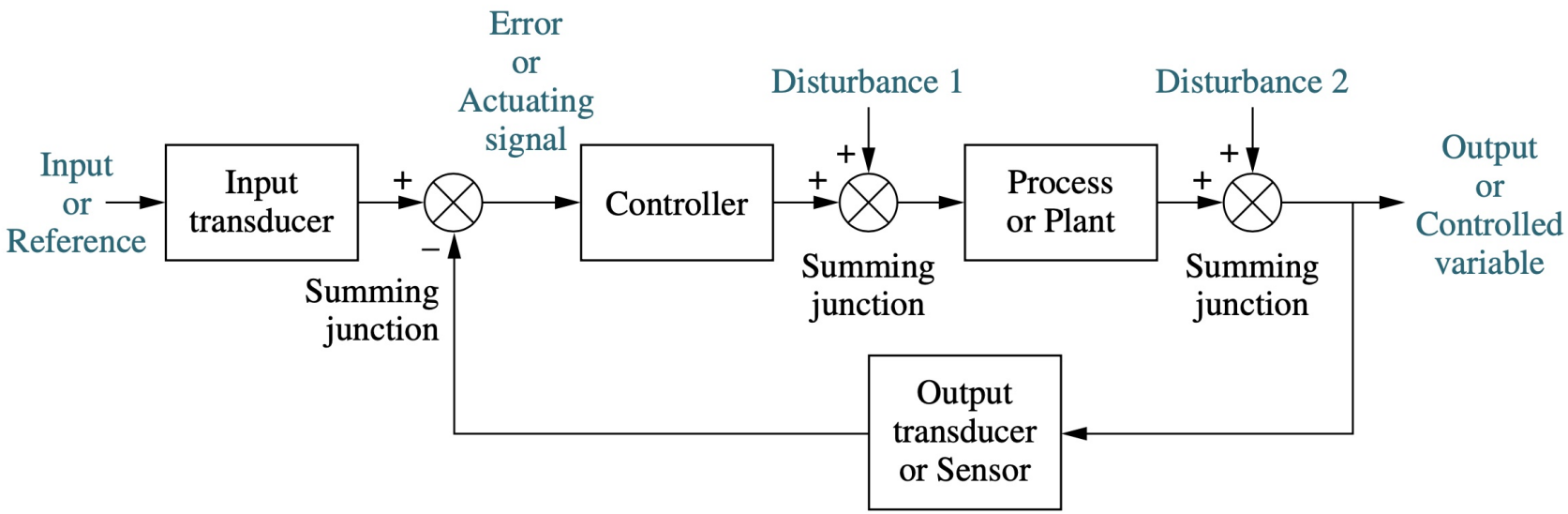
- Transient response: Comfort and patience of passenger
- Steady-state error: Safety and convenience of passenger

Control Theory: Types of Control Systems



Examples: Toaster, spring mass damper system with constant force, study time for an exam to get an AA grade

Sensitivity to disturbances and inability to correct for disturbances



Control Theory: System Components and Aspects

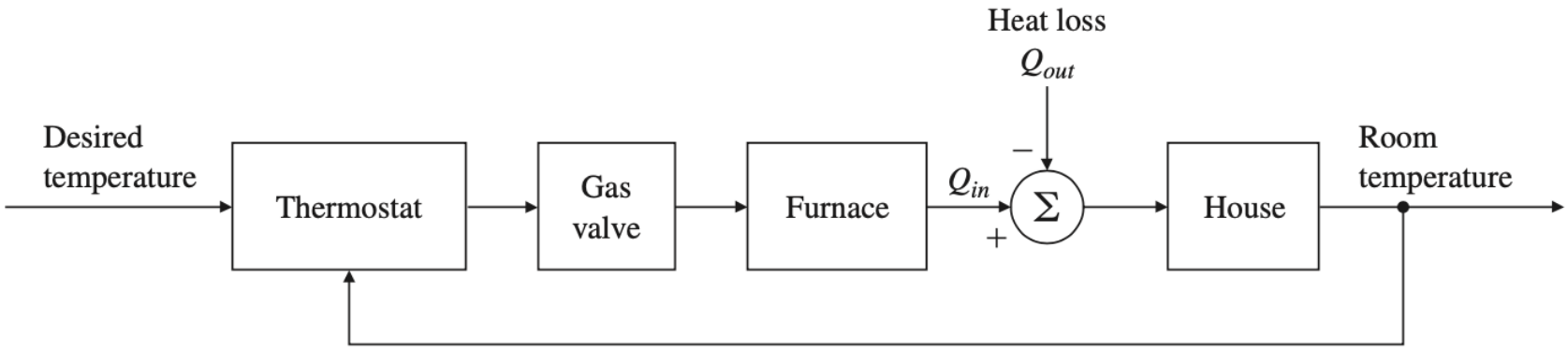
- **Input transducer**: Conversion of the form of the input to that used by the controller.
- **Output transducer (sensor)**: Measures the output response and converts it into the form used by the controller.

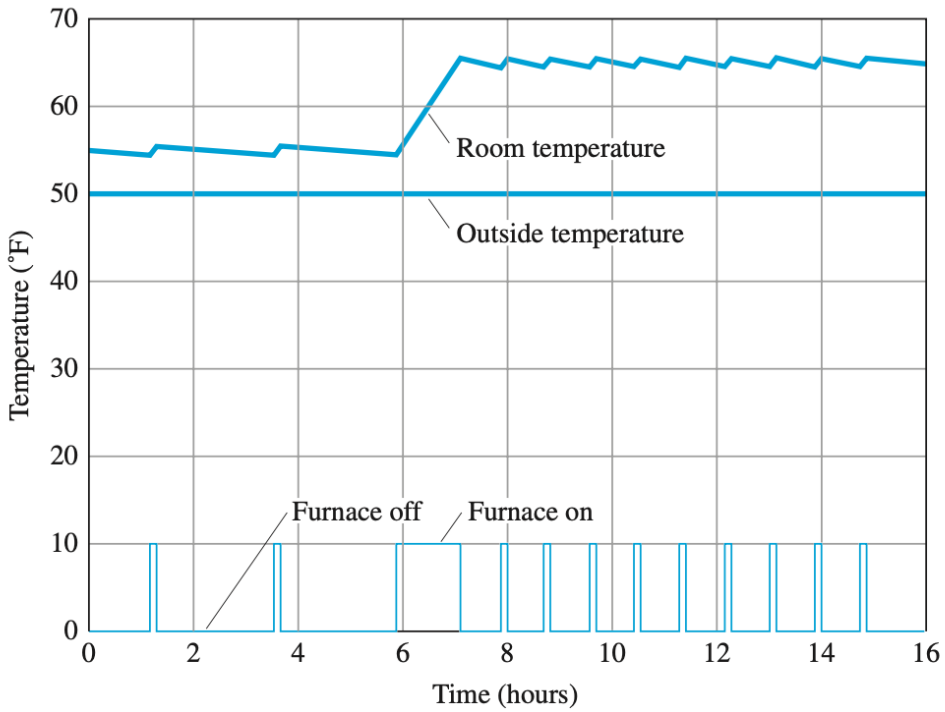
Different aspects of control problem	Example
Plant	Aircraft
Control inputs	Control surface deflections
Disturbance inputs	Wind inputs
Outputs	Vehicle translation and rotation position and velocities
Control goals: Performance and design criteria, Control specification	Speed of achieving commanded output, Handling qualities, Stability

- Input and output are defined w.r.t. a plant.
- Output of one plant can be the input of another plant.
- **Characterization of the controller**: Speed, accuracy, and stability
- Why is feedback important?



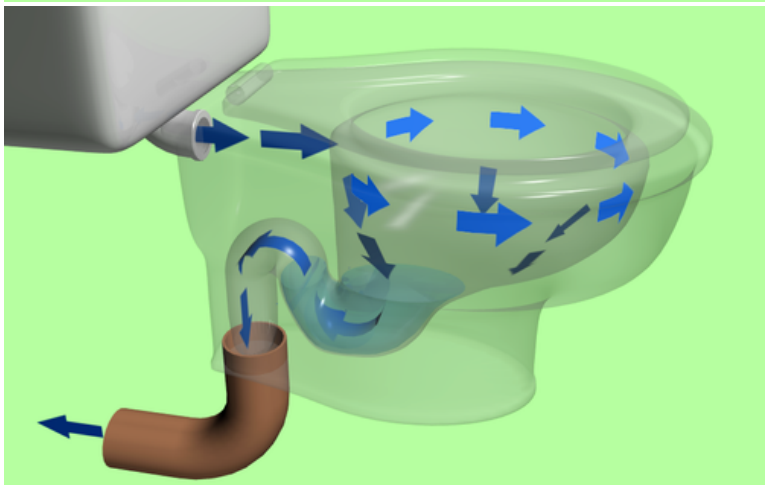
Control Theory: Room Temperature Control System

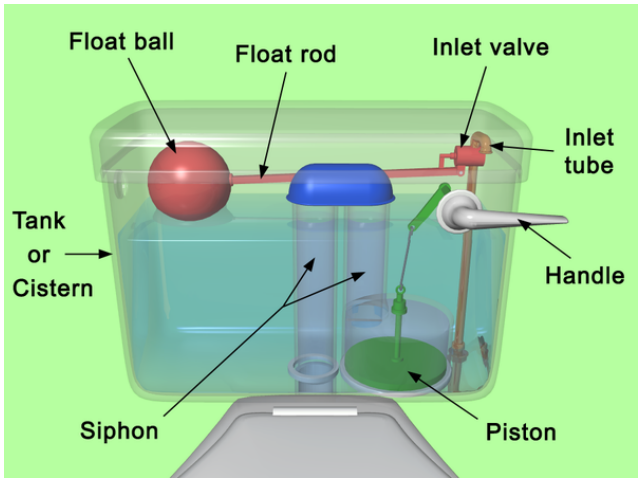


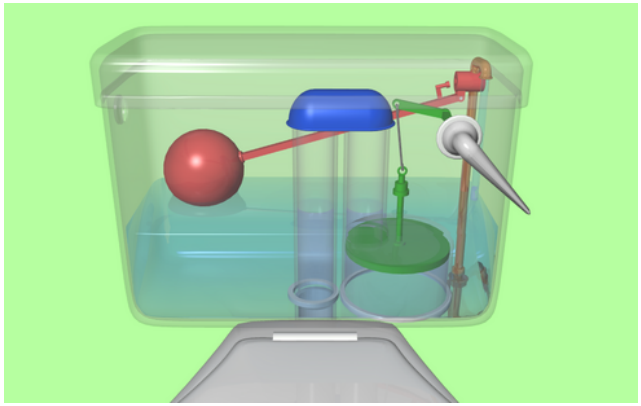


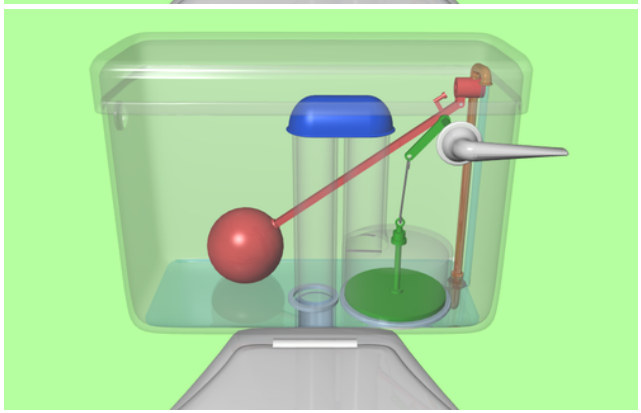
Control Theory: Flushing Mechanism



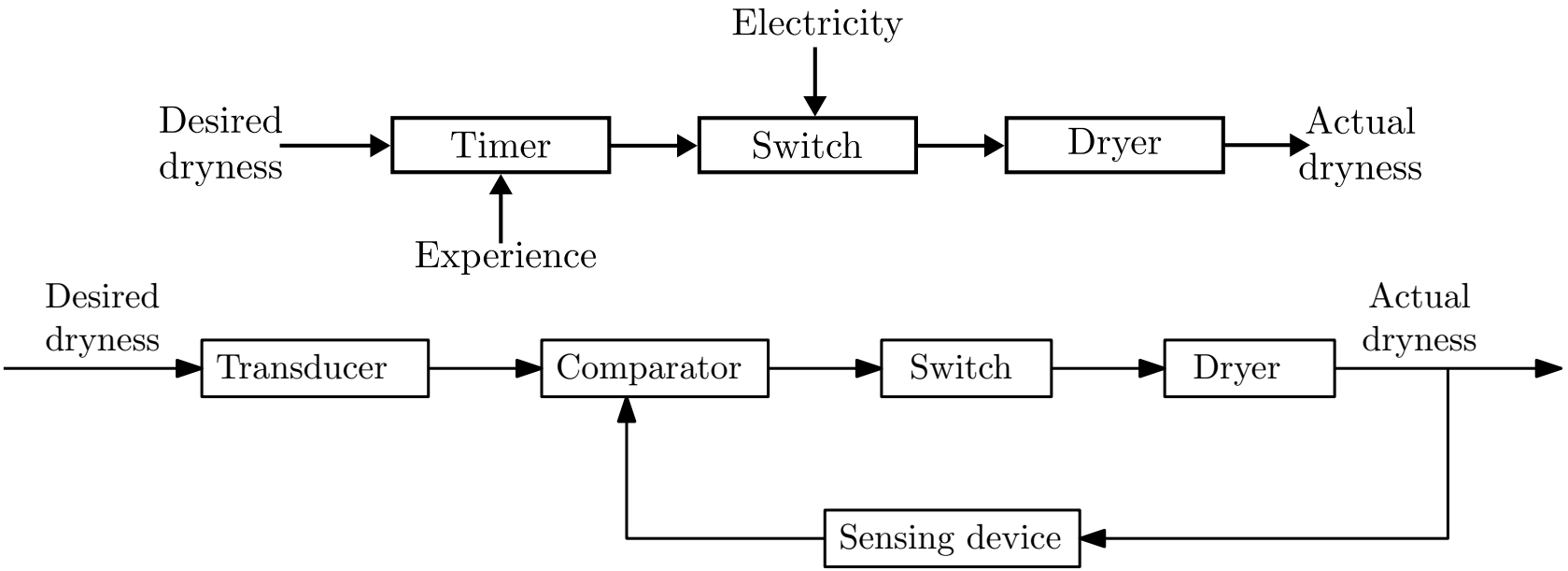


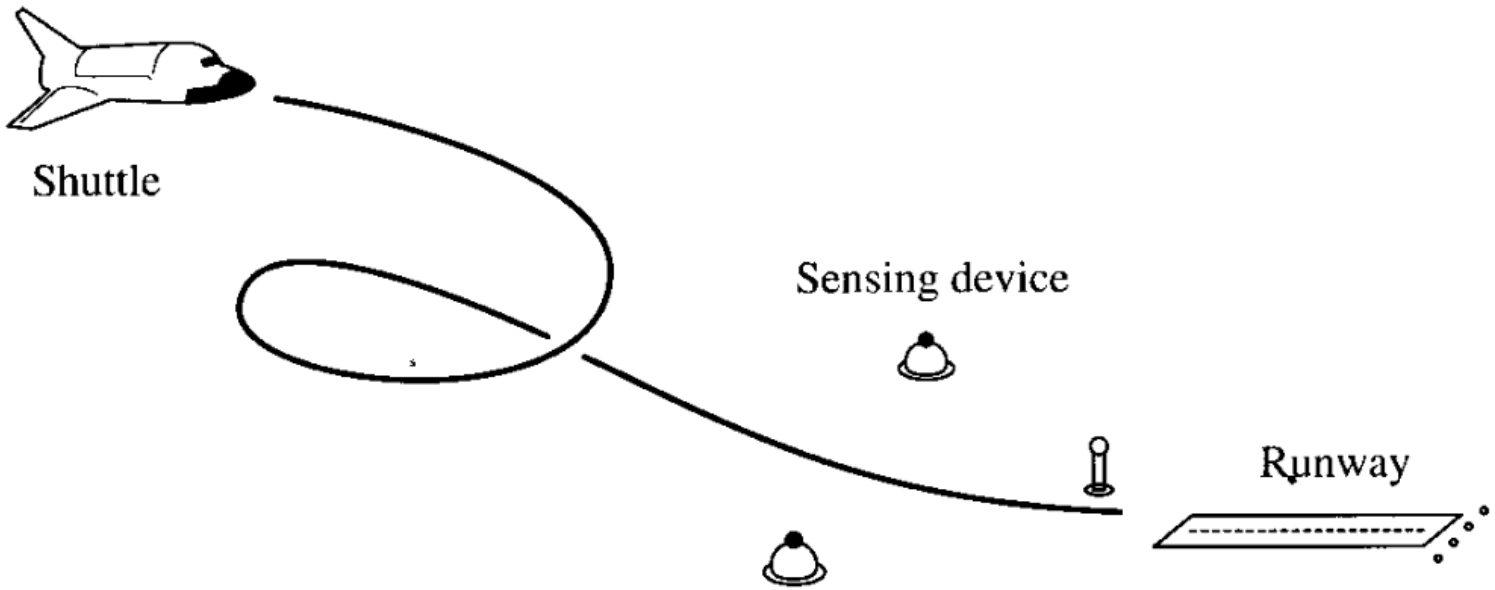






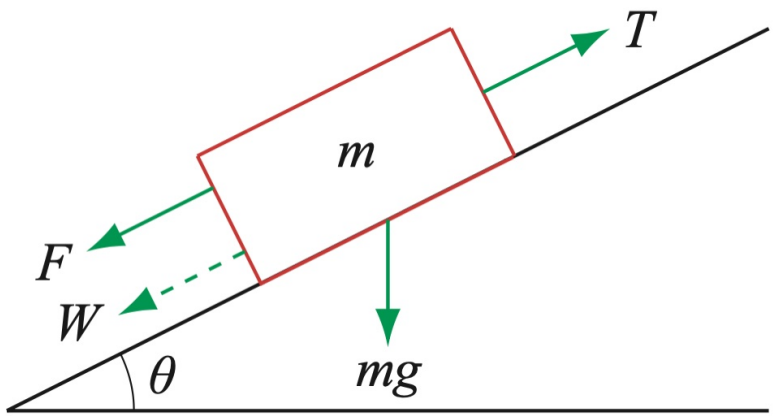
Control Theory: Dryer and Shuttle Landing





Control Theory: Cruise Control

- Mathematical model of vehicle on the road at an angle θ
- Forces acting on the vehicle



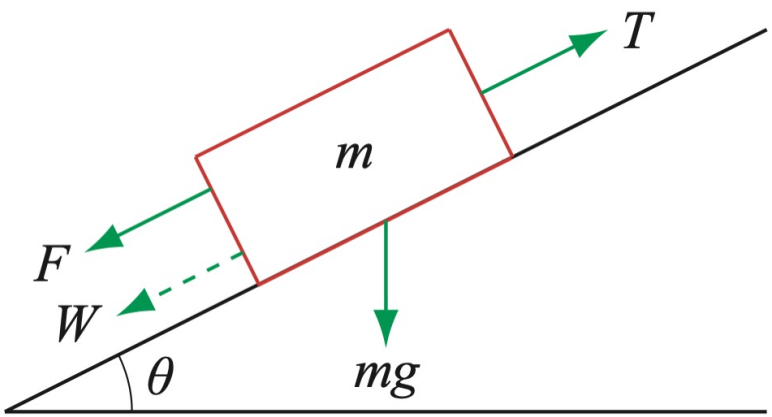
- Equation of motion

$$m\dot{v} = T - F - W = au - bv - mg\sin\theta - D$$

- Known parameter: Engine gain a
- Uncertain mass m, friction coefficient b, and disturbance W.
- **Mass** m: Number of passengers and their masses
- **Friction coefficient** b: Road condition, which may change from one segment of the road to another
- **Disturbance** W: Road angle and the air drag
- Both b and W time-varying during the motion of the vehicle



- How to design the open-loop control?



- Nominal model, with $W = 0$

$$\hat{m}\dot{v} = au - \hat{b}v$$

- Steady-state value, at $\dot{v} = 0$,

$$0 = au - \hat{b}v_{ss} \implies v_{ss} = \frac{au}{\hat{b}}$$

- Constant control to make

$$v_{ss} = v_{\text{des}}$$

$$u = \frac{\hat{b}v_{\text{des}}}{a}$$

- With this control applied to actual system with true parameters (b, m) in presence of disturbance

$$m\dot{v} = \hat{b}v_{\text{des}} - bv - W$$

- With W constant, steady-state

$$v_{ss} = \frac{\hat{b}v_{\text{des}} - W}{b}$$

- Steady-state speed error

$$v_{\text{des}} - v_{ss} = \left(\frac{b - \hat{b}}{b}\right) v_{\text{des}} + \frac{W}{b}$$

Control Theory: Cruise Control

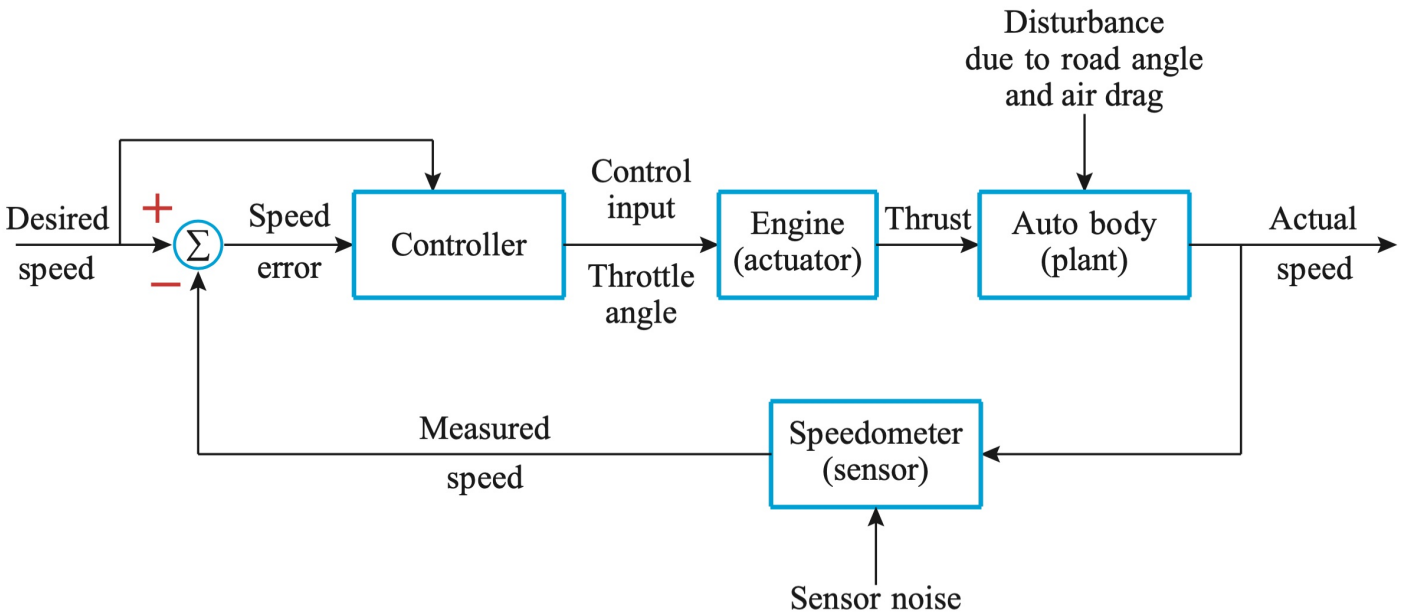
- Closed-loop system, with speed as feedback

$$u = \underbrace{\frac{\hat{b} v_{\text{des}}}{a}}_{\text{Feedforward control}} + \underbrace{K\left(v_{\text{des}} - v\right)}_{\text{Feedback control}}$$

$$m\dot{v} = \hat{b} v_{\text{des}} + aK\left(v_{\text{des}} - v\right) - bv - W = -\left(b + aK\right) v + \left(\hat{b} + aK\right) v_{\text{des}} - W$$

- What would be the steady-state speed error in this case?

$$v_{\text{des}} - v_{ss} = \left(\frac{b - \hat{b}}{b + aK}\right) v_{\text{des}} + \frac{W}{b + aK}$$



Control Theory: Success in Control Engineering

- Plant (process to be controlled)
- Objectives
- Sensors
- Actuators
- Communications
- Computing
- Architectures and interfacing
- Algorithms
- Accounting for disturbances and uncertainty
- Homogeneity

- **Plant**: Physical layout of a plant, physical models
- Knowledge of the basic energy balance, mass balance, and material flows in the system
- **Objectives of control design**:
  - What does one want to achieve?
  - What variables need to be controlled to achieve these objectives?
  - What level of performance is necessary?
- **Sensors**: Eye of the controller
- "If you can measure it, you can control it"
- **Actuators**: Ability to affect or actuate the system to move the process from the current state to a desired state



Control Theory: Success in Control Engineering

- **Communication systems**: Interconnecting sensors to actuators
- Communication delay in transmission: failure of control
- Computational delays, numerical precision, ease of design and implementation
- **Architectures and interfacing**: what to connect to what?
- Centralized control: Complexity, cost, time constraints in computation, maintainability, reliability
- Partition of control problem into manageable sub-systems
- **Algorithms**: Connect the sensors to the actuators (Heart of control engineering)
- Example: Eye-hand coordination for tennis player
- **Disturbances and Uncertainty**: Significant impact on the system performance
- Desired level of performance by using relatively simple models
- **Homogeneity**: Interconnected systems are only as good as their weakest element.
- Focus on the weakest component to get the best return for a given level of investment.



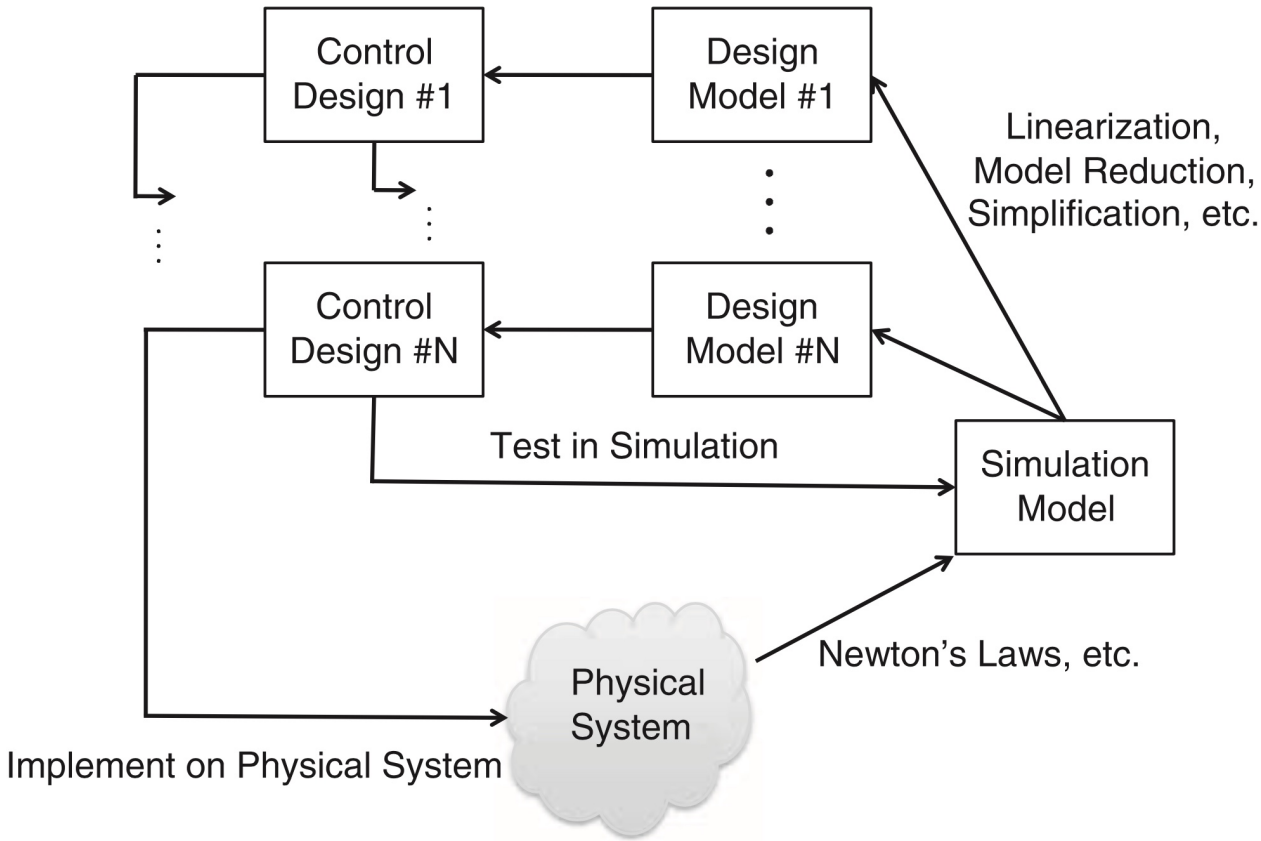
Control Theory: Requirements

- What are the features of a well-designed control system?
- System must be stable.
- **Stability**: Think of it as a requirement mandating that the output of a system should be well-behaved when its input is well-behaved.
- Output must track the desired output (reference or command) with zero or small steady-state error.
- Transient response of the system must be acceptable.
- Effect of disturbance and measurement noise on the system's response should be made as small as possible.
- Design should be robust to model uncertainties.
- Design should meet any constraints associated with the control input.



Control Systems: Design Aspects

- Analytical study of the physical system
  - Modelling
  - Development of mathematical equation description
  - Analysis: quantitative/qualitative
  - Design
- Different models depending on requirements and operating ranges



Laplace and Inverse Laplace Transforms

Laplace transform

Bilateral Laplace transform of a general signal $f(t)$, where $s = \sigma + j\omega$

$$\boxed{\mathcal{L}[f(t)] = \mathcal{F}(s) \triangleq \int_{-\infty}^{\infty} f(t)e^{-st}dt}$$

Unilateral Laplace transform

$$\boxed{\mathcal{L}[f(t)] = \mathcal{F}(s) \triangleq \int_{0_-}^{\infty} f(t)e^{-st}dt}$$

Inverse Laplace transform

$$\boxed{f(t) = \mathcal{L}^{-1}[\mathcal{F}(s)] \triangleq \frac{1}{2\pi j}\int_{\sigma-j\infty}^{\sigma+j\infty} \mathcal{F}(s)e^{st}ds}$$



Laplace Transform of Signals

Signal	Transform	ROC
$\delta(t)$	1	All s
$u(t)$ $[-u(-t)]$	$\frac{1}{s}$	$\Re\{s\} > 0$ $[\Re\{s\} < 0]$
$\frac{t^n}{(n-1)!}u(t)$ $\left[-\frac{t^n}{(n-1)!}u(-t)\right]$	$\frac{1}{s^n}$	$\Re\{s\} > 0$ $[\Re\{s\} < 0]$
$e^{-at}u(t)$ $\left[-e^{-at}u(-t)\right]$	$\frac{1}{s+a}$	$\Re\{s\} > -a$ $[\Re\{s\} < -a]$
$\frac{t^n}{(n-1)!}e^{-at}u(t)$ $\left[-\frac{t^n}{(n-1)!}e^{-at}u(-t)\right]$	$\frac{1}{(s+a)^n}$	$\Re\{s\} > -a$ $[\Re\{s\} < -a]$
$\delta(t-T)$	e^{-sT}	All s
$\cos(\omega_0 t)u(t)$	$\frac{s}{s^2+\omega_0^2}$	$\Re\{s\} > 0$
$\sin(\omega_0 t)u(t)$	$\frac{\omega_0}{s^2+\omega_0^2}$	$\Re\{s\} > 0$
$e^{-at}\cos(\omega_0 t)u(t)$	$\frac{s+a}{(s+a)^2+\omega_0^2}$	$\Re\{s\} > 0$
$e^{-at}\sin(\omega_0 t)u(t)$	$\frac{\omega_0}{(s+a)^2+\omega_0^2}$	$\Re\{s\} > 0$
$u_n(t)$	s^n	All s



Laplace Transforms Properties

Item no.	Theorem	Name
1.	$\mathscr{L}[f(t)] = F(s) = \int_{0-}^{\infty} f(t)e^{-st}dt$	Definition
2.	$\mathscr{L}[kf(t)] = kF(s)$	Linearity theorem
3.	$\mathscr{L}[f_1(t) + f_2(t)] = F_1(s) + F_2(s)$	Linearity theorem
4.	$\mathscr{L}[e^{-at}f(t)] = F(s+a)$	Frequency shift theorem
5.	$\mathscr{L}[f(t-T)] = e^{-sT}F(s)$	Time shift theorem
6.	$\mathscr{L}[f(at)] = \frac{1}{a}F\left(\frac{s}{a}\right)$	Scaling theorem
7.	$\mathscr{L}\left[\frac{df}{dt}\right] = sF(s) - f(0-)$	Differentiation theorem
8.	$\mathscr{L}\left[\frac{d^2f}{dt^2}\right] = s^2F(s) - sf(0-) - f'(0-)$	Differentiation theorem
9.	$\mathscr{L}\left[\frac{d^nf}{dt^n}\right] = s^nF(s) - \sum_{k=1}^{n} s^{n-k}f^{k-1}(0-)$	Differentiation theorem
10.	$\mathscr{L}\left[\int_{0-}^{t} f(\tau)d\tau\right] = \frac{F(s)}{s}$	Integration theorem
11.	$f(\infty) = \lim_{s\to 0} sF(s)$	Final value theorem[1]
12.	$f(0+) = \lim_{s\to\infty} sF(s)$	Initial value theorem[2]

Convolution sum (integral)

$$y(t) = x(t) \circledast h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$



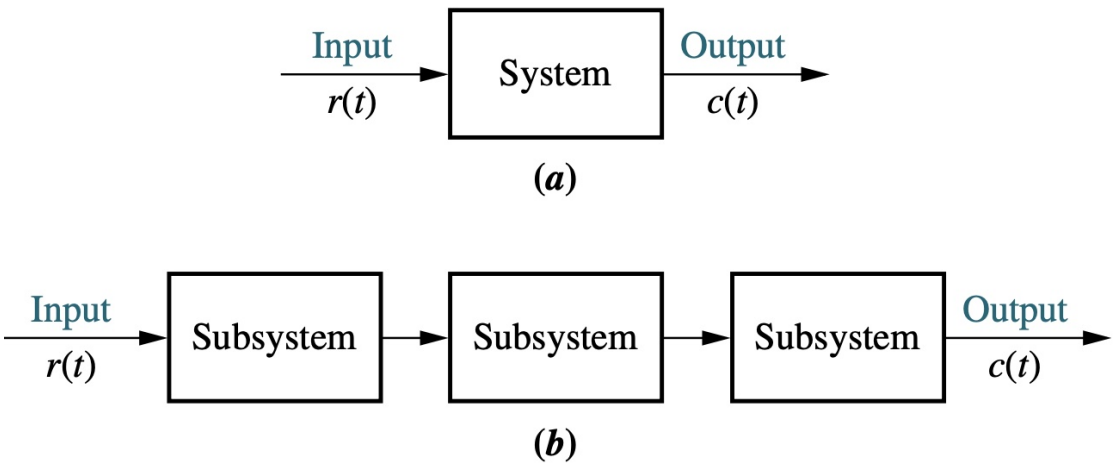
Control System: Transfer Function

- Development of a mathematical model of the system

  $\Longrightarrow$ Frequency domain approach: Transfer functions

  $\Longrightarrow$ Time domain approach: State equations

- Fundamental physical laws of science and engineering
- Ohm's law and Kirchhoff's laws, Newton's laws
- Differential equations relating input, output, and system parameters



- Input, output, and system are distinct and separate parts.
- Interconnection of several sub-systems
- Reference input and controlled variable

Control System: Transfer Function

- Consider a system governed by n^{th}-order LTI differential equation

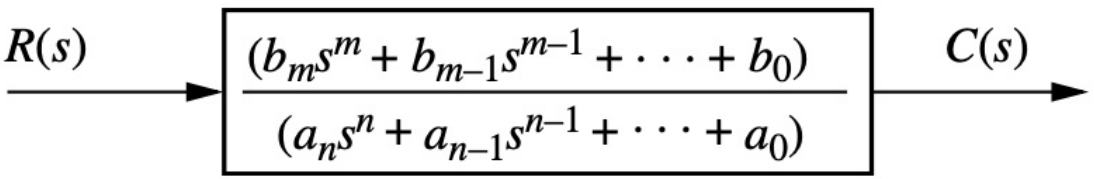
$$a_n \frac{d^n c(t)}{dt^n} + a_{n-1} \frac{d^{n-1} c(t)}{dt^{n-1}} + \cdots + a_0 c(t) = b_m \frac{d^m r(t)}{dt^m} + b_{m-1} \frac{d^{m-1} r(t)}{dt^{m-1}} + \cdots + b_0 r(t)$$

- Taking Laplace transform on both sides

$$a_n s^n C(s) + a_{n-1} s^{n-1} C(s) + \cdots + a_0 C(s) + \text{I.C. terms involving } c(t)$$
$$= b_m s^m R(s) + b_{m-1} s^{m-1} R(s) + \cdots + b_0 R(s) + \text{I.C. terms involving } r(t)$$

- Assuming all I.C. to be zero, **the transfer function** is given by

$$\boxed{G(s) = \frac{C(s)}{R(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + \cdots + b_0}{a_n s^n + a_{n-1} s^{n-1} + \cdots + a_0}} \implies \boxed{C(s) = R(s)G(s)}$$



Why State-Space based Methods?

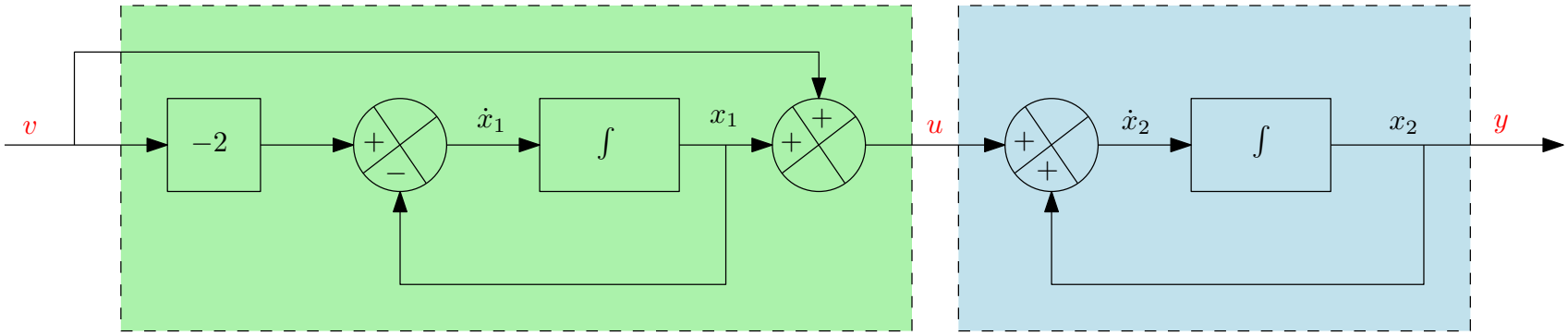
- Transfer function-based approaches
- Various control designs using root-locus or frequency domain techniques
- Why should one go for the state-space-based approach?
- Consider a system and compensator given by

$$G_p(s) = \frac{1}{s-1}, \quad G_c(s) = \frac{s-1}{s+1}, \quad G(s) = G_p(s)G_c(s) = \frac{1}{s+1}$$



- What are the possible difficulties with the above situations?
- Pole-zero cancellation
- Loss of controllability or observability of system

Why State-Space Methods?



$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}-1 & 0\\ 1 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}-2\\ 1\end{bmatrix} v, \quad \begin{bmatrix}x_1(0)\\ x_2(0)\end{bmatrix} = \begin{bmatrix}x_{10}\\ x_{20}\end{bmatrix}, \quad y = \begin{bmatrix}0 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

$$\dot{x}_1 = -x_1 - 2v, \; x_1(0) = x_{10} \implies x_1(t) = e^{-t}x_{10} - 2e^{-t} \circledast v$$

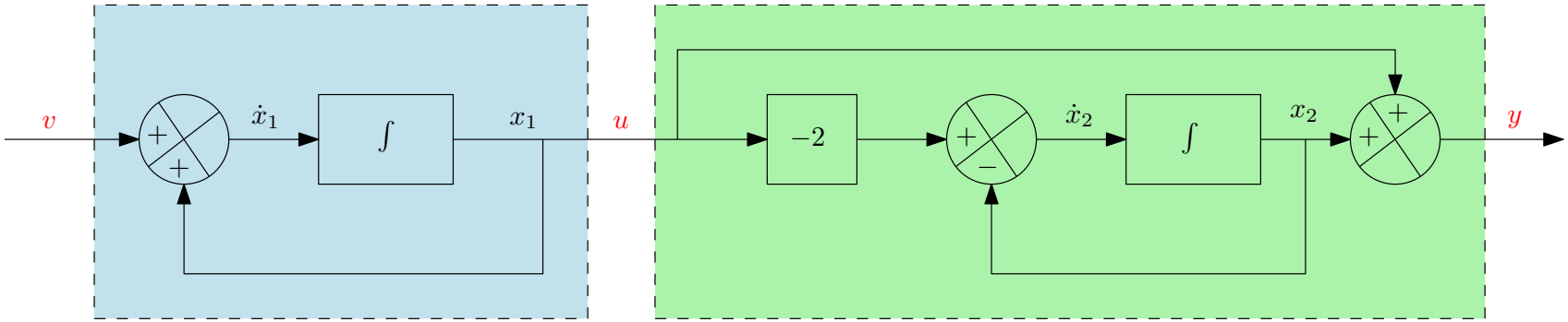
$$Y(s) = X_2(s) = \frac{x_{20}}{s-1} + \frac{x_{10}}{(s-1)(s+1)} + \frac{V(s)}{s+1}$$

$$y(t) = x_2(t) = e^{t}x_{20} + \frac{1}{2}\left(e^{t} - e^{-t}\right)x_{10} + e^{-t} \circledast v$$

- For nonzero initial conditions, $y(t) \to \infty$, except, $\boxed{x_{10} = -2x_{20}}$.
- Is it because of issues of exact pole-zero cancellation? No



Why State-Space Methods?



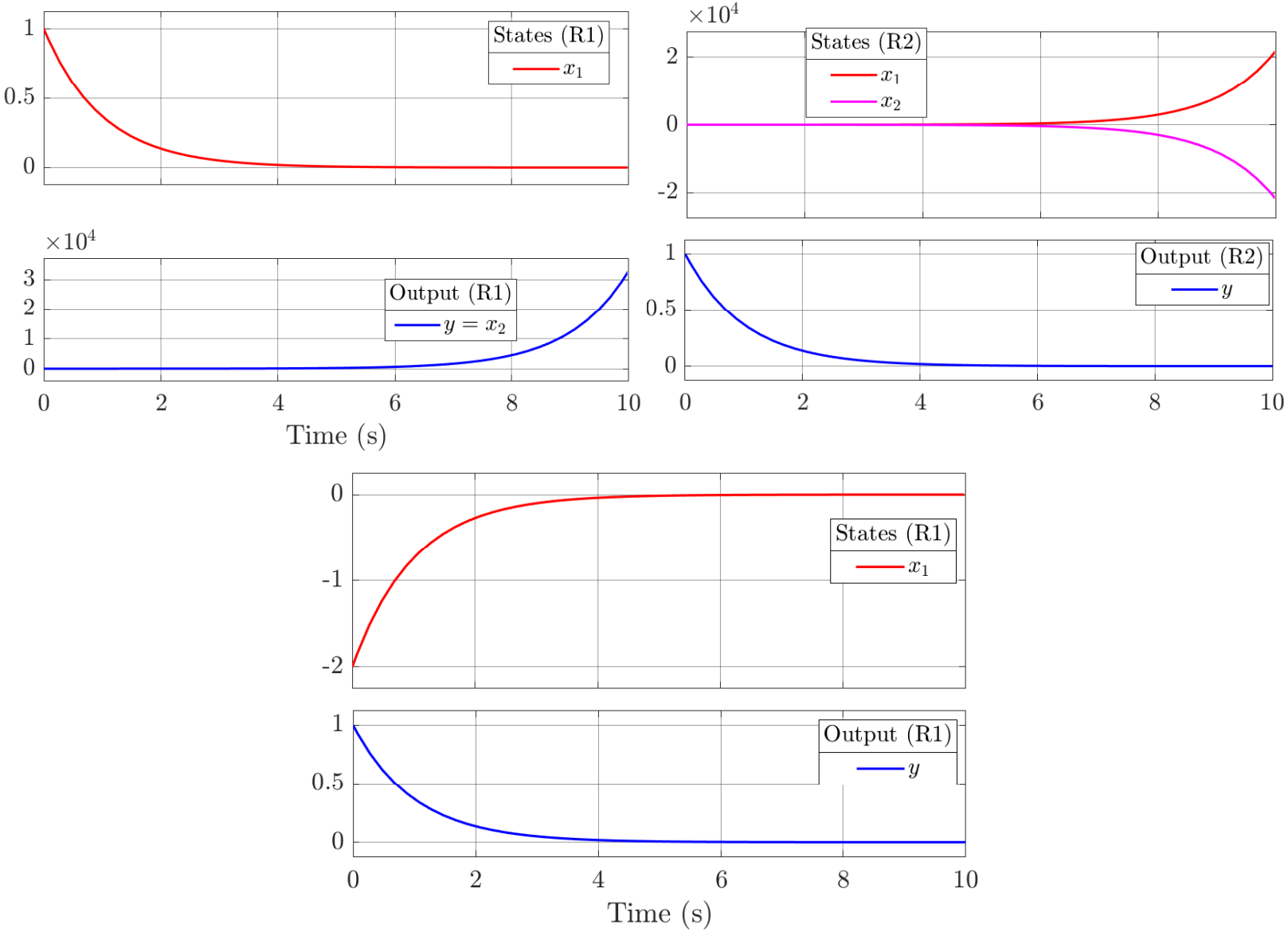
$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}1 & 0\\ -2 & -1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}1\\ 0\end{bmatrix} v, \quad \begin{bmatrix}x_1(0)\\ x_2(0)\end{bmatrix} = \begin{bmatrix}x_{10}\\ x_{20}\end{bmatrix}, \quad y = \begin{bmatrix}1 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

$$y(t) = (x_{10} + x_{20})\, e^{-t} + e^{-t} \circledast v$$

- For nonzero initial conditions, $y(t) \to 0$
- Internally unstable system, with states growing unbounded
- Internal behavior is different and more complicated than external behavior
- **Not all modes appear in the output of the system**
- Transfer function defined under zero initial condition and may not display all modes of the system



Why State-Space Methods? – Demonstration



References

References

- **Nise, Norman S. "Control Systems Engineering", 7th Ed, 2014.**
- Kailath, Thomas, "Linear Systems", Pearson Publication, 2016.
- Goodwin, Graham C.; Graebe, Stefan F.; Salgado, Mario E.; "Control System Design", Prentice Hall, 2000.

